

Sharper bounds for RE-03

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Abstract

For the fixed-partition HODLR approximation problem with $n = 2^L k$, this note develops the bounds

$$c \min \left\{ n, \frac{kL}{\varepsilon} + \frac{k}{\varepsilon^2} \right\} \leq q_*(n, k, \varepsilon) \leq C \min \left\{ n, \frac{kL^2}{\varepsilon} + \frac{kL}{\varepsilon^2} \right\}.$$

The proposed upper bound is obtained by a row-isotropic perturbation inequality for truncated SVD, a noisy Gaussian Nystrom calculation, a constant-accuracy perforated pilot, and a globally shared final regression. The proposed new lower term kL/ε uses weighted hierarchical Stiefel frames, large fixed rank-constrained anchors, and a transcript-divergence argument valid for adaptive queries on both sides. The earlier k/ε^2 lower-bound argument is included in an appendix, so the combined argument can be read without an external unpublished dependency. These bounds still do not match in general. They retain the characterization $q_* = \Theta(n)$ for $\varepsilon \leq \sqrt{k/n}$ and rule out the earlier candidate rate $kL + k/\varepsilon^2$ as a uniform answer.

Status (13 September 2026). This is a partial research manuscript, not a full solution of RE-03. A separate Codex AI agent passed an independent informal audit of the stated partial proofs; see the accompanying `independent-review.md`. No formal verification or external human peer review is asserted. The numerical checks are sanity checks, not proof certificates; the original continuation suite is incomplete in the supplied archive, and only the available assembly smoke checks were rerun. No matching algorithm for the lower expression is supplied.

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1 Problem, principal bounds, and scope

Let $n = 2^L k$, where $k \geq 1$ and $L \geq 2$ are integers, and let $0 < \varepsilon < 1/2$. The class $\mathcal{H}_{n,k}$ consists of matrices whose two ordered off-diagonal sibling blocks at each node of the prescribed balanced binary partition have rank at most k . The terminal diagonal blocks of order k are unrestricted. Set

$$\text{OPT}(A) = \min_{H \in \mathcal{H}_{n,k}} \|A - H\|_F.$$

One query returns Av or $A^\top v$. The side and vector may depend measurably on earlier answers and independent randomness. Only vector queries are charged; a matrix with s query columns costs s calls. The output must be in $\mathcal{H}_{n,k}$ and satisfy

$$\mathbb{P}\{\|A - B\|_F \leq (1 + \varepsilon) \text{OPT}(A)\} \geq 0.99 \quad \text{for every fixed } A \in \mathbb{R}^{n \times n}. \quad (1)$$

This is the quantifier order and exact-real arithmetic model of RE-03.¹

The repository cites an upper bound $O(\min\{n, kL^4/\varepsilon^3\})$ and a lower bound $\Omega(kL + k/\varepsilon)$ in the regime $n \geq c_0 k/\varepsilon$.² The earlier supplied note [3] proposed the stronger accuracy lower bound $c_* \min\{n, k/\varepsilon^2\}$ and an exact-recovery obstruction $k(L + 1)$. Its relevant arguments are preserved in Appendix A and Section 2.

Theorem 1.1 (Combined bounds). *For every admissible (n, k, ε) , the arguments in this manuscript give*

$$c \min\left\{n, \frac{kL}{\varepsilon} + \frac{k}{\varepsilon^2}\right\} \leq q_*(n, k, \varepsilon) \leq C \min\left\{n, \frac{kL^2}{\varepsilon} + \frac{kL}{\varepsilon^2}\right\}, \quad (2)$$

where $c, C > 0$ are universal. One conservative lower constant is

$$D_0 = 2^{24}, \quad c_* = (6400D_0^2)^{-1}, \quad c = c_*/8.$$

The upper bound is nonadaptive and admits the explicit query budget in Theorem 7.3. Independently, $q_* \geq k(L + 1)$.

The constants are deliberately loose. An asymptotic bound with these constants is not a claim that the reference implementation is efficient at moderate matrix sizes. In particular, its theorem-parameter mode normally selects the n -query fallback on small numerical examples.

Corollary 1.2 (Full-recovery regime). *For $0 < \varepsilon \leq \sqrt{k/n}$,*

$$c_* n \leq q_*(n, k, \varepsilon) \leq n.$$

Thus $q_* = \Theta(n)$ in this regime with universal constants.

Corollary 1.3 (A previously plausible smaller rate is excluded). *There is no universal bound $q_* = O(\min\{n, kL + k/\varepsilon^2\})$.*

Proof. Take $k = 1$, $n = 2^L$, and $\varepsilon = L^{-1/2}$ for $L > 4$. The lower expression in (2) is at least a constant times $L^{3/2}$, while $L + \varepsilon^{-2} = 2L$. Both are below n for large L . Their ratio is unbounded. $\square$

The remaining gap is substantive. At fixed positive accuracy and increasing L , the expressions in (2) scale as kL and kL^2 before capping. At $\varepsilon = L^{-1/2}$ they scale as $kL^{3/2}$ and $kL^{5/2}$. Neither the n -query fallback nor the saturation corollary closes these gaps.

¹RE-03, *Optimal matvec query complexity of HODLR approximation*, problem statement and September 10, 2026 audit; source [1].

²Chen et al., *Near-optimal hierarchical matrix approximation from matrix-vector products*, Theorems 1.1–1.2 and Section 6; source [2].

2 Block geometry and the exact baseline

Let $\mathcal{R}$ be the collection of all ordered sibling blocks (I, J) and $\mathcal{D}$ the terminal diagonal blocks. These sets partition the entries. For a rectangular matrix C , let $[C]_k$ be a fixed measurable choice of a best rank-at-most- k SVD truncation and write $\tau_k(C)^2 = \sum_{i>k} \sigma_i(C)^2$.

Lemma 2.1 (Separable optimum). *For every matrix A ,*

$$\text{OPT}(A)^2 = \sum_{(I,J) \in \mathcal{R}} \tau_k(A_{I,J})^2. \quad (3)$$

For any $B \in \mathcal{H}_{n,k}$, each quantity $\|A_{I,J} - B_{I,J}\|_F^2 - \tau_k(A_{I,J})^2$ is nonnegative, as are the diagonal-block errors. Their sum is $\|A - B\|_F^2 - \text{OPT}(A)^2$.

Proof. The squared Frobenius norm is the sum of the squared norms on disjoint blocks. The rank constraints on different sibling blocks are independent. Minimize on each block by an SVD, and copy the unrestricted diagonal blocks. $\square$

Proposition 2.2 (The exact n -query upper bound). *Querying Ae_j for $j = 1, \dots, n$ and then applying the blockwise projection of Lemma 2.1 returns an exactly optimal matrix in $\mathcal{H}_{n,k}$.*

Proposition 2.3 (Adaptive-safe exact-recovery obstruction). *Every algorithm satisfying (1) uses at least $k(L+1)$ queries in the worst case.*

Proof. In each ordered sibling block allow only its first k columns to be nonzero, and allow arbitrary entries in the $k \times k$ diagonal leaves. This is a linear subspace $\mathcal{V} \subseteq \mathcal{H}_{n,k}$. At each of the L levels its sibling blocks contribute nk free entries, and the leaves contribute another nk , so $\dim \mathcal{V} = nk(L+1)$.

Place an independent standard Gaussian prior on coordinates in an orthonormal basis of $\mathcal{V}$. Conditional on the algorithm's seed and previous answers, a new vector query supplies at most n linear scalar observations. By induction, the unobserved component remains a nondegenerate Gaussian on a subspace of dimension at least $nk(L+1) - nq$. Adaptivity does not add constraints beyond those answers: the next linear measurement is fixed once the past is fixed. If $q < k(L+1)$, the posterior has no atoms, and any measurable point estimate has conditional probability zero of exact recovery.

But $\text{OPT}(A) = 0$ on this prior, so (1) requires exact recovery with probability at least 0.99 for each fixed input and hence on average. This contradicts the preceding posterior statement. $\square$

3 A perturbation bound for rank truncation

The following deterministic inequality is the main ingredient of the improved upper bound. It separates the effect of noise in the leading left singular space from the full Frobenius size of the noise.

Lemma 3.1 (Deterministic truncation inequality). *Let $X, E \in \mathbb{R}^{r \times m}$ and $1 \leq k \leq \min(r, m)$. Let P be a rank- k projector onto leading left singular vectors of X , set $T = (I - P)X$ and $\tau = \|T\|_F$. Then*

$$\|X - [X + E]_k\|_F^2 - \tau^2 \leq 4\tau \|PE\|_F + 6\|E\|_F^2 + 4\sqrt{2k} \|T^\top E\|_F. \quad (4)$$

No independence, spectral gap, or distributional hypothesis is needed.

Proof. Put $F = PE$, $G = (I - P)E$, $Z = X + F$, and $f = \|F\|_F$. Let Q and Q_0 be rank- k right singular projectors for $Z + G$ and Z , respectively. The optimality of Q gives

$$\|Z(I - Q)\|_F^2 \leq \tau_k(Z)^2 + a, \quad a = 2\sqrt{2k} \|T^\top G\|_F + \|G\|_F^2.$$

Indeed, expand $(Z + G)^\top(Z + G)$ in the variational characterization of its leading right singular subspace. The cross term is bounded using $Z^\top G = T^\top G$ and $\|Q - Q_0\|_F \leq \sqrt{2k}$. The remaining term obeys $\text{tr}(G^\top G(Q - Q_0)) \leq \|G\|_F^2$.

Since $\tau_k(Z) \leq \tau + f$,

$$\begin{aligned} \|X(I - Q)\|_F^2 &\leq (\sqrt{(\tau + f)^2 + a} + f)^2 \\ &\leq \tau^2 + 4\tau f + 5f^2 + 2a. \end{aligned}$$

The second line follows from $\sqrt{x^2 + a} \leq x + \sqrt{a}$ and $2f\sqrt{a} \leq f^2 + a$. Also $[X + E]_k = (X + E)Q$, and right-orthogonality gives

$$\|X - (X + E)Q\|_F^2 = \|X(I - Q)\|_F^2 + \|EQ\|_F^2.$$

Finally $T^\top G = T^\top E$ and $5\|F\|_F^2 + 2\|G\|_F^2 + \|E\|_F^2 \leq 6\|E\|_F^2$. $\square$

Corollary 3.2 (Row-isotropic noise). *Suppose the second moments of E satisfy*

$$\mathbb{E}[E_{i\alpha}E_{j\beta}] = h\delta_{ij}\Sigma_{\alpha\beta}, \quad \Sigma \succeq 0,$$

and put $S = \text{tr } \Sigma$. Then

$$\mathbb{E}\|X - [X + E]_k\|_F^2 \leq \tau_k(X)^2 + 10\tau_k(X)\sqrt{khS} + 6rhS. \quad (5)$$

Proof. The three second moments needed in Lemma 3.1 are

$$\mathbb{E}\|PE\|_F^2 = khS, \quad \mathbb{E}\|T^\top E\|_F^2 = h\|T\|_F^2 S, \quad \mathbb{E}\|E\|_F^2 = rhS.$$

Apply Cauchy-Schwarz to the two unsquared norms and use $4 + 4\sqrt{2} < 10$. If $r < k$, truncate at r instead: the risk is simply rhS , and the upper expression with $\tau_k(X) = 0$ remains valid. $\square$

Lemma 3.3 (Gaussian least-squares moments). *Let $G \in \mathbb{R}^{t \times r}$ be standard Gaussian, $t > r + 1$, and let $F \in \mathbb{R}^{t \times m}$ be independent of G with independent centered Gaussian rows of covariance Σ . Then $E = G^\dagger F$ has second moments*

$$\mathbb{E}[E_{i\alpha}E_{j\beta}] = \frac{\delta_{ij}\Sigma_{\alpha\beta}}{t - r - 1}.$$

Proof. Conditional on G , the row covariance factor is $(G^\top G)^{-1}$. Sign changes of individual columns of G make the expected off-diagonal entries of this inverse zero. A Schur complement expresses each diagonal entry as the reciprocal squared norm of one Gaussian column projected onto the orthogonal complement of the others. That squared norm is χ_{t-r+1}^2 . Integrating its density gives $\mathbb{E}[(\chi_\nu^2)^{-1}] = 1/(\nu - 2)$ for $\nu > 2$. Thus $\mathbb{E}(G^\top G)^{-1} = I/(t - r - 1)$, proving the claim. $\square$

4 Noisy Gaussian Nyström approximation

Consider a fixed block $C \in \mathbb{R}^{m_1 \times m_2}$ with additive sketch contaminants $M \in \mathbb{R}^{m_1 \times a}$ and $N \in \mathbb{R}^{b \times m_2}$. Draw independent standard Gaussian matrices of the following sizes:

$$\Omega : m_2 \times s, \quad \Gamma : a \times s, \quad \Psi : m_1 \times t, \quad \Theta : b \times t.$$

Form

$$Q = \text{orth}(C\Omega + M\Gamma), \quad \hat{X} = (\Psi^\top Q)^\dagger (\Psi^\top C + \Theta^\top N), \quad \hat{C} = Q[\hat{X}]_k. \quad (6)$$

The number of columns of Q may be less than s .

Theorem 4.1 (Noisy block estimate). *Assume $s > k + 1$ and $t > s + 1$. Set*

$$\tau = \tau_k(C), \quad \alpha = \frac{k}{s - k - 1}, \quad d = t - s - 1, \quad D = (1 + \alpha)\tau^2 + \alpha \|M\|_F^2 + \|N\|_F^2.$$

Then the construction (6) satisfies

$$\begin{aligned} \mathbb{E} \|C - \hat{C}\|_F^2 &\leq \tau^2 + \alpha(\tau^2 + \|M\|_F^2) \\ &\quad + 10\tau \sqrt{\frac{kD}{d}} + \frac{6sD}{d}. \end{aligned} \quad (7)$$

The estimate is rank at most k .

Proof. If $k \geq \min(m_1, m_2)$, the same arguments below can use a smaller true rank in the right-space calculation; alternatively use its rank- k SVD with zero singular directions completed orthogonally. We give the case $k \leq \min(m_1, m_2)$ explicitly.

Write $C = C_k + C_{\text{tail}}$ using an SVD and let V_1 contain its leading k right singular vectors. Let $\Omega_1 = V_1^\top \Omega$ and complete V_1 to an orthogonal basis. Because Ω_1 has full row rank almost surely, multiplication of the range sketch by $\Omega_1^\dagger$ gives

$$\|(I - QQ^\top)C_k\|_F^2 \leq \|(C_{\text{tail}}\Omega + M\Gamma)\Omega_1^\dagger\|_F^2.$$

The Gaussian coordinates entering $C_{\text{tail}}\Omega$ and Γ are independent of Ω_1 . Their cross term has expectation zero. The Gaussian pseudoinverse calculation of Lemma 3.3, transposed, (a standard ingredient in Gaussian range finding [4]), therefore gives

$$\mathbb{E} \|(I - QQ^\top)C_k\|_F^2 \leq \alpha(\tau^2 + \|M\|_F^2). \quad (8)$$

One may verify the omitted right isometry in the first display by multiplying on the right by $V_1^\top$; it does not change the Frobenius norm.

The ideal projected rank- k approximation satisfies

$$\begin{aligned} \|C - Q[Q^\top C]_k\|_F^2 &\leq \|C - QQ^\top C_k\|_F^2 \\ &= \tau^2 + \|(I - QQ^\top)C_k\|_F^2. \end{aligned} \quad (9)$$

The equality follows from orthogonality of the right singular spaces of C_k and C_{tail} . In particular, for $C_\perp = (I - QQ^\top)C$,

$$\mathbb{E} \|C_\perp\|_F^2 \leq (1 + \alpha)\tau^2 + \alpha \|M\|_F^2.$$

Condition on Q and let r be its number of columns. The Gaussian matrix $\Psi^\top Q$ is independent of $\Psi^\top C_\perp$. Consequently

$$\hat{X} = Q^\top C + E, \quad E = (\Psi^\top Q)^\dagger (\Psi^\top C_\perp + \Theta^\top N)$$

has the second moments of Lemma 3.3 with $\Sigma = C_\perp^\top C_\perp + N^\top N$ and $h = 1/(t - r - 1)$. Apply Corollary 3.2 with $X = Q^\top C$. Its tail is at most τ , because projection is a contraction and $Q^\top C_k$ is an admissible rank- k competitor. The error outside Q is orthogonal to the error inside Q . Use (9), $r \leq s$, $t - r - 1 \geq d$, and Jensen's inequality on the square root after averaging over Q . This gives (7). If $r = 0$, the output is zero and the same bound follows from (9); no inverse is needed. $\square$

5 Perforated peeling with a global error bound

Only the first $h = L - 1$ levels need nontrivial rank truncation. The remaining size- $2k$ diagonal blocks are unrestricted in $\mathcal{H}_{n,k}$, since their $k \times k$ sibling blocks already have rank at most k .

5.1 Sketches and query accounting

At a level $\ell \in \{1, \dots, h\}$, the target blocks have order $m = n/2^\ell$. Treat the two orientations separately. For the upper orientation let (I_b, J_b) be the left/right children of parent b . Let $H_{<\ell}$ contain the approximations of the strictly coarser sibling blocks, with zero entries elsewhere, and write $R = A - H_{<\ell}$. Draw a uniform independent hash $f(b) \in \{1, \dots, B\}$ and independent Gaussian $m \times s$ matrices Ω_b . For each bucket a , query the matrix whose restriction to J_b is Ω_b when $f(b) = a$, and is zero elsewhere. Subtract the known product with $H_{<\ell}$. The rows I_b of the bucket $f(b)$ give

$$Y_b = A_{I_b, J_b} \Omega_b + \sum_{c \neq b: f(c)=f(b)} R_{I_b, J_c} \Omega_c. \quad (10)$$

This uses at most Bs vector queries for one orientation.

Independently draw a left hash g and Gaussian $m \times t$ matrices Ψ_b . Query $A^\top$ on bucketed vectors supported on the I_b and subtract $H_{<\ell}^\top$. Transpose the appropriate outputs to get

$$Z_b = \Psi_b^\top A_{I_b, J_b} + \sum_{c \neq b: g(c)=g(b)} \Psi_c^\top R_{I_c, J_b}. \quad (11)$$

This costs at most Bt queries. Return $Q_b[(\Psi_b^\top Q_b)^\dagger Z_b]_k$ on this block, where $Q_b = \text{orth}(Y_b)$. The lower orientation exchanges the two children and uses fresh independent randomness. Freeze $H_{<\ell}$ until both orientations are complete.

Every queried vector depends only on random sketches, hashes, n , and k ; subtraction of earlier approximations occurs after the oracle responses. Thus all actual query vectors may be chosen in advance. Empty buckets may be skipped. The per-level cost is at most $2B(s + t)$.

5.2 Contamination energies

Let

$$a_\ell = \sum_{\text{ordered blocks at level } \ell} \tau_k(A_{I,J})^2, \quad O = \sum_{\ell=1}^h a_\ell = \text{OPT}(A)^2.$$

Let $T_{\ell-1}$ be the squared error on the strictly coarser blocks. Condition on the past. For a right sketch the contaminant M_b is the horizontal concatenation of the coarser residual blocks that

collide with b ; for a left sketch N_b is their vertical analogue. Concatenated Gaussian matrices put (10)–(11) exactly in the setting of Theorem 4.1.

Every pair of distinct parents collides with probability $1/B$. Moreover, the corresponding residual entries belong to strictly coarser blocks. Across both orientations each such entry is counted at most once in the right-contamination sum, and at most once in the left-contamination sum. Therefore, with the expectation here only over hashes,

$$\mathbb{E} \sum_b \|M_b\|_F^2 \leq T_{\ell-1}/B, \quad \mathbb{E} \sum_b \|N_b\|_F^2 \leq T_{\ell-1}/B. \quad (12)$$

The two sums need not be independent. No independence between different output blocks is needed in the analysis.

5.3 Summing the errors before choosing parameters

Put

$$\alpha = \frac{k}{s-k-1}, \quad \beta = \frac{s}{t-s-1}, \quad \gamma = \sqrt{\frac{k}{t-s-1}},$$

$$r_0 = \alpha + 6\beta(1+\alpha), \quad c_0 = 10\gamma\sqrt{1+\alpha}.$$

Summing Theorem 4.1 over all targets at this level, using (12), and applying Cauchy–Schwarz to the mixed terms gives

$$\mathbb{E}[\text{level-}\ell \text{ error}^2 \mid \text{past}] \leq (1 + r_0 + c_0)a_\ell + \frac{r_0}{B}T_{\ell-1} + \frac{c_0}{\sqrt{B}}\sqrt{a_\ell T_{\ell-1}}. \quad (13)$$

For clarity, the total quantity D in the mixed term of the block theorem has conditional expectation at most $(1+\alpha)(a_\ell + T_{\ell-1}/B)$. Use $\sqrt{a(a+x)} \leq a + \sqrt{ax}$. The coefficient on $T_{\ell-1}/B$ from the non-mixed terms is no greater than r_0 , giving (13).

Choose $B = h$ and let S be the expected squared error on all nonterminal sibling blocks. Since $T_{\ell-1}$ is a sum of earlier nonnegative block errors, $\sum_\ell \mathbb{E}T_{\ell-1} \leq hS$. Also

$$\sum_\ell \mathbb{E}\sqrt{a_\ell T_{\ell-1}} \leq \sqrt{O \sum_\ell \mathbb{E}T_{\ell-1}} \leq \sqrt{OhS}.$$

Consequently

$$S \leq (1 + r_0 + c_0)O + r_0S + c_0\sqrt{OS}. \quad (14)$$

All expectations are finite: Theorem 4.1 and induction on the finite number of levels establish this before the summation argument.

5.4 Parameters and the last diagonal fit

Set

$$\eta = \varepsilon/2600, \quad s = k + 1 + \lceil k/\eta \rceil, \quad t = s + 1 + \lceil s/\eta \rceil, \quad B = h. \quad (15)$$

Then $\alpha \leq \eta$, $\beta \leq \eta$, and $\gamma \leq \eta$. The last inequality follows from $s \geq k/\eta$. Since $\eta < 1/100$, we have $r_0 \leq 8\eta$ and $c_0 \leq 11\eta$. Using $\sqrt{OS} \leq (O + S)/2$ in (14) gives

$$S \leq \frac{1 + r_0 + \frac{3}{2}c_0}{1 - r_0 - \frac{1}{2}c_0}O \leq (1 + 50\eta)O.$$

The same inequality gives $S = 0$ when $O = 0$; division by O is unnecessary.

For the final size- $2k$ diagonal blocks, choose

$$p = 2k + 1 + \lceil 2k/\eta \rceil \quad (16)$$

and query $A\Omega$ for an independent standard Gaussian $\Omega \in \mathbb{R}^{n \times p}$. Subtract the now-known off-diagonal approximation $H_{\text{off}}\Omega$. On a diagonal block I , set

$$\hat{D}_I = ((A - H_{\text{off}})\Omega)_I \Omega_I^\dagger.$$

The contaminating columns outside I are independent of Ω_I . The same Gaussian inverse moment calculation gives conditional total diagonal error at most $2k/(p - 2k - 1)$ times the off-diagonal error. Thus the final squared error obeys

$$\mathbb{E} \|A - \hat{H}\|_{\text{F}}^2 \leq (1 + \eta)(1 + 50\eta)O \leq (1 + 52\eta)O. \quad (17)$$

Every output block meets its rank constraint, so $\hat{H} \in \mathcal{H}_{n,k}$.

Theorem 5.1 (Explicit nonadaptive upper bound). *With (15)–(16), the preceding construction uses at most*

$$Q = 2(L - 1)^2(s + t) + p \quad (18)$$

queries and satisfies (1). Selecting the n -query exact baseline whenever $Q \geq n$ gives

$$q_* \leq \min\{n, Q\} = O\left(\min\{n, kL^2/\varepsilon^2\}\right).$$

Proof. The query count follows directly from the two orientations, buckets, sketch widths, and last diagonal fit. If $O > 0$, the excess $Z = \|A - \hat{H}\|_{\text{F}}^2 - O$ is nonnegative by Lemma 2.1. Equation (17) and Markov's inequality give

$$\mathbb{P}\{Z > (2\varepsilon + \varepsilon^2)O\} \leq \frac{52\eta}{2\varepsilon + \varepsilon^2} \leq 0.01.$$

If $O = 0$, (17) says the output error is zero almost surely. Finally $s = O(k/\eta)$, $t = O(k/\eta^2)$, and $p = O(k/\eta)$, with constants independent of n, k, L, ε . $\square$

6 Joint fitting when the column spaces are supplied

The repeated coefficient-fitting cost across levels is avoidable when the correct block column spaces are supplied. This conditional result does not supply those spaces and therefore is not an upper bound for the general problem by itself.

For each ordered sibling block (I, J) , fix an orthonormal matrix $U_{I,J} \in \mathbb{R}^{|I| \times r_{I,J}}$ with $0 \leq r_{I,J} \leq k$. Let $\mathcal{V}(U)$ be the linear subspace of matrices whose block (I, J) has columns in $\text{col}(U_{I,J})$, with arbitrary diagonal leaves. Then $\mathcal{V}(U) \subseteq \mathcal{H}_{n,k}$. For a column index j , concatenate the corresponding embedded sibling bases along its root-to-leaf path and the coordinate basis of its leaf. Call the resulting matrix Z_j . The row supports of its constituent pieces are disjoint, so Z_j is orthonormal and

$$r_j := \#\text{columns}(Z_j) = k + \sum_{(I,J): j \in J} r_{I,J} \leq k(L + 1) =: r_{\max}^{\text{bound}}.$$

Put $r_{\max} = \max_j r_j$ and $A_U[:, j] = Z_j Z_j^\top A[:, j]$. This is the Frobenius projection of A onto $\mathcal{V}(U)$.

Draw a fresh standard Gaussian $\Omega \in \mathbb{R}^{n \times q}$ and use q transpose queries to obtain $Y = A^\top \Omega$. Define

$$B[:, j] = Z_j(\Omega^\top Z_j)^\dagger Y[j, :]^\top. \quad (19)$$

The same q responses are used for every column and every level.

Proposition 6.1 (Known-space coefficient fit). *If $q > r_{\max} + 1$, the fit (19) belongs to $\mathcal{V}(U)$ and satisfies the exact expectation identity*

$$\mathbb{E} \|A - B\|_F^2 = \sum_{j=1}^n \left(1 + \frac{r_j}{q - r_j - 1}\right) \|(I - Z_j Z_j^\top) A[:, j]\|^2. \quad (20)$$

In particular, taking

$$q = r_{\max} + 1 + \lceil 50r_{\max}/\varepsilon \rceil$$

gives $\|A - B\|_F \leq (1 + \varepsilon) \|A - A_U\|_F$ with probability at least 0.99, using $O(kL/\varepsilon)$ queries.

Proof. Write $A[:, j] = Z_j c_j + e_j$ with $Z_j^\top e_j = 0$. For each fixed column, $G_j = \Omega^\top Z_j$ is standard Gaussian and independent of $\Omega^\top e_j$, whose entries are independent $N(0, \|e_j\|^2)$. The least-squares coefficient error is $G_j^\dagger \Omega^\top e_j$. Lemma 3.3 gives its expected squared norm as $r_j \|e_j\|^2 / (q - r_j - 1)$. Orthogonality to e_j proves the identity for that column, and linearity of expectation proves (20). The fits for different columns need not be independent.

The error in excess of $\|A - A_U\|_F^2$ is nonnegative and has mean at most $(\varepsilon/50) \|A - A_U\|_F^2$ for the stated width. Markov's inequality bounds the failure probability by $1/(50(2 + \varepsilon)) < 0.01$. If the projection error is zero, the expectation identity gives exact recovery almost surely. $\square$

If the spaces are obtained using earlier queries, the proposition remains valid conditionally when Ω is independent of those earlier queries. If the supplied bases contain the true leading k left singular vectors of each block, then $\|A - A_U\|_F = \text{OPT}(A)$. Neither observation yields those bases from the permitted oracle at no cost. The accompanying test helper that forms bases from a full SVD is expressly privileged input preparation, not part of a claimed query algorithm.

This result isolates one missing step more precisely: the kL/ε scale is sufficient for fitting all coefficients once suitable rank- k spaces are available. The present work has no uniformly accurate procedure for learning such spaces at the total cost needed to match the combined lower bound.

7 A two-stage improvement with shared regression

The coefficient accuracy needed during peeling is lower than the accuracy needed in the final answer. Keeping the larger range spaces, using only a constant-factor pilot fit, and then refitting all blocks simultaneously gives a stronger upper bound than Theorem 5.1.

7.1 A constant-accuracy pilot and accurate retained ranges

Let $h = L - 1$ and choose

$$\eta = \varepsilon/500, \quad s = k + 1 + \lceil k/\eta \rceil, \quad t = s + 1 + 1000s, \quad B = h. \quad (21)$$

Run the peeling construction of Section 5 on the first h levels with these parameters, but do not perform its final diagonal fit. Retain every range basis Q_b as well as the rank- k pilot block

used for subtraction at later levels. Write $r_b = \dim \text{col}(Q_b) \leq s$. The two orientations at a level still use a frozen pilot from coarser levels.

In the notation of (13), $\alpha \leq \eta < 10^{-3}$, $\beta = 10^{-3}$, and $\gamma \leq \sqrt{\eta/1000}$. Hence $r_0 < 0.01$ and $c_0 < 0.02$. The global inequality (14) gives, for the expected nonterminal pilot error S ,

$$S \leq \frac{1 + r_0 + \frac{3}{2}c_0}{1 - r_0 - \frac{1}{2}c_0} O < 2O, \quad O = \text{OPT}(A)^2. \quad (22)$$

This is a constant-accuracy statement about the pilot, not the final output.

For each target block C_b , define the retained-range miss

$$M_b^{\text{miss}} = \left\| (I - Q_b Q_b^\top) [C_b]_k \right\|_{\text{F}}^2.$$

The right-space estimate (8), conditioned on the past and then averaged over hashes, implies

$$\mathbb{E} \sum_b M_b^{\text{miss}} \leq \alpha \left(O + \frac{1}{h} \sum_{\ell=1}^h \mathbb{E} T_{\ell-1} \right) \leq \eta(O + S) \leq 3\eta O. \quad (23)$$

Let I_Q denote the squared Frobenius error of the best rank- k approximation within each retained column space, with exact size- $2k$ diagonal blocks. By (9),

$$\mathbb{E} I_Q \leq (1 + 3\eta) O. \quad (24)$$

7.2 Shared regression in larger spaces

Now condition on the entire pilot. For each column j , form an orthonormal matrix Z_j by concatenating its size- $2k$ diagonal coordinate basis and all retained Q_b on sibling row intervals for which j belongs to the column interval. The row supports are disjoint. Thus

$$r_j = \dim \text{col}(Z_j) \leq 2k + sh =: r_{\text{bd}}.$$

Let $A_Q[:, j] = Z_j Z_j^\top A[:, j]$, $e_j = A[:, j] - A_Q[:, j]$, and $R_Q^2 = \sum_j \|e_j\|^2$. This larger linear model has no rank- k restriction on the retained coefficients. In particular,

$$R_Q^2 \leq I_Q. \quad (25)$$

Draw a fresh standard Gaussian $\Omega \in \mathbb{R}^{n \times p}$ and obtain $A^\top \Omega$ with p transpose queries. Fit each column by $Z_j (\Omega^\top Z_j)^\dagger \Omega^\top A[:, j]$. On every nonterminal sibling block, truncate this fitted block to rank k . Leave the fitted size- $2k$ diagonal blocks unchanged. Call the resulting proper HODLR matrix $\hat{H}$.

Lemma 7.1 (Row isotropy survives shared fitting). *Fix a retained block $b = (I, J)$ with $r = r_b > 0$, and write its fitted coefficient matrix before truncation as $X_b + E_b$, where $X_b = Q_b^\top A_{I,J}$. For $p > \max_j r_j + 1$, there is a positive semidefinite matrix Σ_b such that*

$$\mathbb{E}[(E_b)_{i\alpha}(E_b)_{j\beta} \mid \text{pilot}] = \delta_{ij}(\Sigma_b)_{\alpha\beta}, \quad \text{tr} \Sigma_b = \sum_{a \in J} \frac{\|e_a\|^2}{p - r_a - 1}.$$

The expectation is over the fresh shared Gaussian sketch. Errors for different columns or blocks are not claimed to be independent.

Proof. Embed Q_b into $\mathbb{R}^n$ on rows I . For every $a \in J$, Z_a contains this same basis. Its other columns have row support outside I , and e_a is orthogonal to the embedded Q_b . Therefore $G = \Omega^\top Q_b$ is independent of all the other design components and all noise vectors $\Omega^\top e_a$ needed for these columns, simultaneously for $a \in J$.

For any fixed orthogonal $O_r \in \mathbb{R}^{r \times r}$, replacing G by GO_r leaves that joint Gaussian distribution invariant. The least-squares coefficient-error block is then replaced by $O_r^\top E_b$, because the transformation acts on these design coordinates and not on the residual noise. Thus E_b has a left-orthogonally invariant distribution. All second moments are finite by Lemma 3.3. Sign changes of one row make cross-row second moments zero, and row permutations make within-row covariance matrices equal. This proves the claimed form with a positive semidefinite Σ_b .

For a fixed column a , the entire design $\Omega^\top Z_a$ is standard Gaussian and independent of $\Omega^\top e_a$. Lemma 3.3 says that every coefficient coordinate has variance $\|e_a\|^2 / (p - r_a - 1)$. Summing these diagonal entries over $a \in J$ proves the trace identity. $\square$

Proposition 7.2 (Conditional proper-fit error). *If $p \geq r_{\text{bd}} + 1 + d$ for $d > 0$, then*

$$\mathbb{E}[\|A - \hat{H}\|_{\text{F}}^2 \mid \text{pilot}] \leq I_Q + 10\sqrt{\frac{kh}{d}}\sqrt{OR_Q^2} + \frac{6sh + 2k}{d}R_Q^2. \quad (26)$$

Proof. Use Lemma 7.1 in Corollary 3.2 for each block, and add the orthogonal error outside its retained space. The coefficient tail $\tau_k(X_b)$ is no larger than $\tau_k(C_b)$. Put $S_b = \text{tr } \Sigma_b$. Every column belongs to exactly one ordered sibling column interval at each of the h nonterminal levels. Hence

$$\sum_b S_b \leq \frac{h}{d} \sum_j \|e_j\|^2 = \frac{h}{d} R_Q^2.$$

Cauchy–Schwarz gives $\sum_b \tau_k(C_b) \sqrt{k S_b} \leq \sqrt{khOR_Q^2/d}$, and $\sum_b r_b S_b \leq shR_Q^2/d$. If $r_b < k$, use the nontruncated case of Corollary 3.2; its stated upper bound remains valid. Zero-dimensional blocks contribute no regression noise.

For a column j , its $2k$ diagonal coefficients have total expected squared error $2k \|e_j\|^2 / (p - r_j - 1)$. Summing these errors adds at most $2kR_Q^2/d$. All block errors add on disjoint entries, proving the result. $\square$

7.3 The improved query bound

Take

$$d = \left\lceil \max \left\{ \frac{100kh}{\eta^2}, \frac{6sh + 2k}{\eta} \right\} \right\rceil, \quad p = 2k + sh + 1 + d. \quad (27)$$

Combining (26), Jensen’s inequality, (25), and (24) gives

$$\mathbb{E} \|A - \hat{H}\|_{\text{F}}^2 \leq \left(1 + 3\eta + \eta\sqrt{1 + 3\eta} + \eta(1 + 3\eta)\right) O \leq (1 + 6\eta)O.$$

If $O = 0$, the expected error is zero. Otherwise the final output is in $\mathcal{H}_{n,k}$, so its excess over O is nonnegative. Markov’s inequality yields failure probability at most $6\eta/(2\varepsilon + \varepsilon^2) \leq 0.006 < 0.01$.

Theorem 7.3 (Two-stage nonadaptive upper bound). *The parameters (21) and (27) give an algorithm satisfying (1) with at most*

$$Q_2 = 2(L - 1)^2(s + t) + p \quad (28)$$

queries. Using exact reconstruction when $Q_2 \geq n$ gives

$$q_*(n, k, \varepsilon) \leq \min\{n, Q_2\} = O\left(\min\left\{n, \frac{kL^2}{\varepsilon} + \frac{kL}{\varepsilon^2}\right\}\right).$$

All query vectors may be drawn in advance. In particular, the final shared Gaussian sketch has a width bounded using r_{bd} , not a width selected from the observed pilot ranks.

Proof. The pilot uses at most $2hB(s+t) = 2h^2(s+t)$ queries and the shared fit uses p transpose queries. Here $s, t = O(k/\varepsilon)$ and $p = O(kL/\varepsilon^2)$. The probability bound was established above. The pilot's query vectors are nonadaptive by the construction in Section 5; the shared sketch is independent of all of them. No oracle products with individual blocks are treated as free operations. $\square$

The extra factor compared with the combined lower bound is now only one factor of depth: before taking the n -query cap, the expression in this upper bound is exactly $L(kL/\varepsilon + k/\varepsilon^2)$. A matching bound without this factor is still not proved.

8 Information in adaptive two-sided Gaussian queries

The next lower-bound construction is nonsymmetric and uses independent Gaussian entries rather than GOE. Both query types must therefore be included in the information calculation.

Lemma 8.1 (Two-sided transcript divergence). *Let $A_0, S \in \mathbb{R}^{n \times n}$ be fixed with $\|S\|_{\text{op}} \leq 1$ and let G have independent $N(0, 1)$ entries. For any randomized adaptive algorithm using at most q queries, let $\mathbb{P}_S$ and $\mathbb{P}_0$ be its transcript laws on $A_0 + \delta S + G$ and $A_0 + G$. Include the seed in the transcript. Then*

$$D_{\text{KL}}(\mathbb{P}_S \| \mathbb{P}_0) \leq \delta^2 q / 2. \quad (29)$$

Proof. Keep separate orthonormal spans V of forward query vectors and U of transpose query vectors. A component already in the appropriate span has a known image; remove it and normalize the remainder. Redundant queries contribute no new divergence.

For a fixed value of S and a fixed history, the observations reveal GV and $U^\top G$. The unobserved component is $P_{U^\perp} G P_{V^\perp}$. It remains a centered isotropic Gaussian on this rectangular subspace. This is proved inductively: conditioned on the past, the next direction is fixed, and the newly exposed row or column of the remaining Gaussian rectangle is independent of the smaller rectangle left unexposed. The same argument works for measurable adaptive choices because they impose no condition beyond the answers already in the history.

For a fresh unit forward vector $v \perp V$, project the output onto $U^\perp$. The omitted U component is already determined by the known transpose answers. At a common history the two conditional means of the new information differ by $\delta P_{U^\perp} S v$, and the conditional covariance is the identity on $U^\perp$. Its one-step divergence is at most $\delta^2/2$. A fresh transpose query similarly has mean difference $\delta P_{V^\perp} S^\top u$ and the same bound. Apply the chain rule and then average over the seed. Early stopping can be padded or treated by zero-contribution steps. $\square$

Remark 8.2 (Known diagonal scaling). Let D be any fixed invertible diagonal matrix and write $A = A'D$. An algorithm for A can be simulated using the same number of queries to A' : a forward query uses $A'(Dv)$, and a transpose query returns $D(A'^\top v)$. Thus the same transcript-divergence bound applies to algorithms run on $A = (A_0 + \delta S + G)D$. The approximation loss for this scaled family must still be analyzed separately; diagonal scaling does not preserve the Frobenius objective. Neither the size of D nor the size of A_0 incurs query cost.

9 A weighted hierarchical frame packing

Assume $L \geq 6$ and put $h = L - 4$. In this section the symbol h is a lower-bound construction parameter, not the upper algorithm's $L - 1$. Use only upper sibling blocks at levels $\ell = 1, \dots, h$. Their orders are $m_\ell = n/2^\ell$, and the smallest is $16k$. For a target block (I_b, J_b) , let C_b be the first k coordinate indices of J_b . These column sets are disjoint across the entire hierarchy: an ancestor's selected indices lie in the leftmost leaf of its right child, and hence in no strictly descendant right child.

Choose an orthonormal $m_\ell \times k$ frame U_b and put it in rows I_b and columns C_b of a square matrix S , setting every other entry to zero. Construct these frames from fine levels to coarse levels. At a block with row interval I_b and order m , require its frame to be orthogonal to all previously constructed frame columns supported inside I_b . There are exactly

$$k(2^{h-\ell} - 1) = m/16 - k$$

such descendant columns. Thus the available subspace has dimension $15m/16 + k$; even the last column of the new frame is chosen in dimension at least $15m/16 + 1$.

All nonzero columns of S are consequently orthonormal. In particular, $\|S\|_{\text{op}} = 1$ and $S \in \mathcal{H}_{n,k}$. Set the fixed diagonal weights to

$$D_{jj}^2 = \begin{cases} m_\ell/n, & j \in C_b \text{ for a level-}\ell \text{ target,} \\ 1/n, & j \text{ is not selected.} \end{cases} \quad (30)$$

The weights do not depend on the chosen frames. At every target level, $\sum_b \|U_b D_{C_b}\|_{\text{F}}^2 = k/2$, so

$$\|SD\|_{\text{F}}^2 = kh/2. \quad (31)$$

Lemma 9.1 (Hierarchical packing). *There is a finite family $\{S_\theta : \theta \in \mathcal{T}\}$ constructed above such that, with $N = |\mathcal{T}|$,*

$$\log N \geq nkh/128, \quad \|(S_\theta - S_{\theta'})D\|_{\text{F}}^2 > kh/128 \quad (\theta \neq \theta'). \quad (32)$$

Here and below logarithms used in information bounds are natural.

Proof. Sample each frame uniformly in its allowed orthogonal complement, conditionally on finer frames. We bound the measure of a small ball around any fixed frame family T .

A newly sampled unit column u lies on a sphere in dimension $d \geq 15m/16 + 1$. For any fixed unit vector v in the ambient space, $\|u - v\| \leq 1/4$ implies $\langle u, v \rangle \geq 31/32$. Projecting v into the allowed subspace cannot enlarge this cap probability. For a standard Gaussian representation of the sphere, put $a = 63/961 < 1/15$ and apply the exponential Markov bound with parameter 2:

$$\mathbb{P}\{u_1 \geq 31/32\} \leq (1 - 4a)^{-1/2} 5^{-(d-1)/2} \leq 2 \cdot 5^{-(d-1)/2} \leq e^{-m/2}.$$

The last inequality holds for $m \geq 16$. This bound remains valid after any previous columns and all finer frames have been fixed.

A level has $M = n/(2m)$ frames. If its weighted squared distance to T is at most $k/64$, its Mk column distances have total square at most $Mk/32$. At least half these columns must therefore have distance at most $1/4$. A union bound over subsets of columns gives the uniform conditional bound

$$\mathbb{P}\{\text{this level has weighted distance}^2 \leq k/64 \mid \text{finer frames}\} \leq 2^{Mk} e^{-mMk/4} \leq e^{-nk/16}.$$

If the total weighted squared distance is at most $kh/128$, at least half of the h levels have distance at most $k/64$. For any specified collection of levels, integrate their conditional bounds from coarse to fine (each is uniform in its finer conditioning). A union bound over level subsets yields

$$\mathbb{P}\{\|(S - T)D\|_F^2 \leq kh/128\} \leq 2^h e^{-nkh/32} \leq e^{-nkh/64}. \quad (33)$$

The last inequality uses $nk \geq 64$, valid for $L \geq 6$.

A greedy construction can therefore choose at least $\lfloor e^{nkh/64} \rfloor$ centers with pairwise distance greater than $\sqrt{kh}/128$: before that many choices, the union of the forbidden balls has measure less than one. Since $nkh/64 \geq 2$, the logarithm of this packing size is at least $nkh/128$. $\square$

10 Anchors and the depth–accuracy lower bound

10.1 A deterministic anchor inequality

Lemma 10.1 (Selected-column loss). *Write $C = [X \ Y]$ with $X \in \mathbb{R}^{m \times k}$ and $\sigma_{\min}(X) \geq s_0 > 0$. Let $B = [B_1 \ B_2]$ have rank at most k . If $\|X - B_1\|_F \leq K$ and $s_0 \geq 2K$, then*

$$\|C - B\|_F^2 - \tau_k(C)^2 \geq \|X - B_1\|_F^2 - \frac{2K}{s_0} \|Y\|_F^2. \quad (34)$$

Proof. The perturbation bound on the smallest singular value makes B_1 full column rank. Thus every column of B_2 belongs to $\text{col}(B_1)$. Let P_X, P_B be the projections onto $\text{col}(X)$ and $\text{col}(B_1)$. The competitor $[X \ P_X Y]$ has rank at most k , while the best possible fit of Y using $\text{col}(B_1)$ is $P_B Y$. It follows that the left side of (34) is at least

$$\|X - B_1\|_F^2 + \text{tr}(Y^\top (P_X - P_B) Y).$$

For equal-dimensional subspaces, $\|P_X - P_B\|_{\text{op}} = \|(I - P_B)P_X\|_{\text{op}}$. Since $(I - P_B)X = (I - P_B)(X - B_1)$, this is at most K/s_0 , and in particular at most $2K/s_0$. Substitution proves the displayed inequality. $\square$

10.2 The hard input distribution

For the core argument assume

$$h = L - 4 \geq 4, \quad \varepsilon h \geq 1. \quad (35)$$

Choose S_θ uniformly from Lemma 9.1. Let G be an independent standard iid Gaussian square matrix. For each target choose a fixed orthonormal frame U_b^0 and construct A_0 by putting $M_0 U_b^0$ in rows I_b and selected columns C_b , with zeros elsewhere. Set

$$A = (A_0 + \delta S_\theta + G)D, \quad \delta^2 = 2^{18} \varepsilon n. \quad (36)$$

The large number M_0 is fixed below, independently of θ, G . Since every clean target block has at most k nonzero columns, $(A_0 + \delta S_\theta)D \in \mathcal{H}_{n,k}$. Hence

$$\text{OPT}(A)^2 \leq \|GD\|_F^2. \quad (37)$$

There are $K_{\text{col}} = k(2^h - 1) = n/16 - k$ selected columns. From (30),

$$\mathbb{E} \|GD\|_F^2 = n \text{tr} D^2 = n \left(\frac{kh}{2} + \frac{n - K_{\text{col}}}{n} \right) \leq nkh.$$

On the selected entries of target blocks, the expected noise energy is

$$\sum_b \mathbb{E} \|(GD)_{I_b, C_b}\|_F^2 = \frac{k}{2} \sum_{\ell=1}^h m_\ell = \frac{kn}{2} (1 - 2^{-h}) < kn/2. \quad (38)$$

Markov's inequality and a union bound therefore give an event of probability at least 0.8 on which

$$\|GD\|_F^2 \leq N_0 := 10nkh, \quad \sum_b \|(GD)_{I_b, C_b}\|_F^2 \leq 5nk. \quad (39)$$

Put

$$K_0 = \frac{3}{2}\sqrt{N_0}, \quad s_0 = 2K_0/\varepsilon, \quad w_{\min} = \sqrt{16k/n}, \quad M_0 = \delta + \frac{s_0 + \sqrt{N_0}}{w_{\min}} + 1.$$

On (39), the selected columns $X_b = A_{I_b, C_b}$ have

$$\sigma_{\min}(X_b) \geq (M_0 - \delta)\sqrt{m_\ell/n} - \sqrt{N_0} \geq s_0.$$

The unselected columns Y_b in each target block contain noise only.

10.3 Decoding from a successful approximation

Let an algorithm satisfying (1) return B . On its success event and (39), every selected-column error is at most K_0 , because the entire error is at most $(1 + \varepsilon)\sqrt{N_0} < K_0$. Apply Lemma 10.1 to each target block. Its remainder coefficient $2K_0/s_0$ equals ε . By nonnegativity of all other local excesses,

$$\begin{aligned} \sum_b \|X_b - B_{I_b, C_b}\|_F^2 &\leq (\|A - B\|_F^2 - \text{OPT}(A)^2) + \varepsilon \sum_b \|Y_b\|_F^2 \\ &\leq (2\varepsilon + \varepsilon^2)N_0 + \varepsilon N_0 \leq 35\varepsilon nkh. \end{aligned} \quad (40)$$

Subtract the fixed anchor on these selected entries and set all other entries to zero. This gives an estimate $\widehat{W}$ of $\delta S_\theta D$ with

$$\begin{aligned} \|\widehat{W} - \delta S_\theta D\|_F^2 &\leq 2 \sum_b \|X_b - B_{I_b, C_b}\|_F^2 + 2 \sum_b \|(GD)_{I_b, C_b}\|_F^2 \\ &\leq 70\varepsilon nkh + 10nk \leq 80\varepsilon nkh. \end{aligned} \quad (41)$$

The last inequality uses $\varepsilon h \geq 1$. The packing's half-separation squared is larger than

$$\delta^2 kh/512 = 512\varepsilon nkh > 80\varepsilon nkh.$$

Thus nearest-center decoding of $\widehat{W}$ identifies θ on that joint event. The average decoding error is at most 0.21: the noise event fails with probability at most 0.2, and the algorithm fails with probability at most 0.01 after averaging its pointwise guarantee over the hard inputs.

10.4 The information contradiction

Known scaling by D does not change the query count. Lemma 8.1 therefore gives, for the full transcript T ,

$$I(\theta; \mathsf{T}) \leq \delta^2 q/2.$$

To see this directly, average the divergences to the common baseline $A_0 + G$. That average equals $I(\theta; \mathsf{T})$ plus the nonnegative divergence of the mixture transcript law to the baseline.

For a uniform label in a family of size N , every decoder obeys

$$p_e \geq 1 - \frac{I(\theta; \mathsf{T}) + \log 2}{\log N}. \quad (42)$$

This follows by conditioning on its error indicator: $H(\theta \mid \mathsf{T}) \leq \log 2 + p_e \log N$. If

$$q \leq \frac{kh}{1024 \cdot 2^{18}\varepsilon},$$

then $I(\theta; \mathsf{T}) \leq nkh/2048 \leq (\log N)/16$. In the core regime $\log N \geq nkh/128 \geq 8$, so (42) gives $p_e > 0.85$, contradicting $p_e \leq 0.21$.

Proposition 10.2 (Depth-accuracy lower bound in the core). *Under (35),*

$$q_*(n, k, \varepsilon) \geq c_d \frac{kh}{\varepsilon}, \quad c_d = (1024 \cdot 2^{18})^{-1}.$$

11 Uniform extension and what is still missing

Appendix A proves, using the earlier supplied argument,

$$q_*(n, k, \varepsilon) \geq c_* \min\{n, k/\varepsilon^2\}, \quad c_* = (6400 \cdot 2^{48})^{-1}. \quad (43)$$

We now remove the auxiliary restrictions from the new depth bound.

In the core, $h \geq L/2$ and $kL/\varepsilon \leq kLh \leq 2kh^2 \leq n$ for $h \geq 4$. Thus Proposition 10.2 implies $q_* \geq (c_d/2) \min\{n, kL/\varepsilon\}$. Outside the core, either $L < 8$, in which case $\varepsilon L < 4$, or $L \geq 8$ and $\varepsilon h < 1$, in which case $\varepsilon L < 1 + 4\varepsilon < 3$. In both cases

$$kL/\varepsilon \leq 4k/\varepsilon^2.$$

Equation (43) therefore yields the depth bound with constant $c_*/4$. This is smaller than $c_d/2$ and works in all cases.

For nonnegative a, b , one has $\min(n, a) + \min(n, b) \geq \min(n, a + b)$. Taking the maximum of the two established lower bounds and using half their sum gives the left side of (2) with $c = c_*/8$. The upper side follows from Theorem 7.3. This completes the detailed argument for Theorem 1.1.

11.1 The roles of the two hard distributions

The depth construction estimates many coefficients spread over scales. Its weighted packing has entropy proportional to nkh , and large fixed anchors convert small global excess into accurate estimation of selected columns. The diagonal weighting keeps the information per whitened query controlled while assigning comparable clean signal energy to each level.

The accuracy construction in the appendix is different. It places nearly degenerate rank choices inside size- $2k$ sibling blocks. The relevant local loss is linear in a small spectral gap, while transcript information is quadratic in that gap. This produces k/ε^2 rather than k/ε . The two constructions cannot simply be multiplied: the packing dimension, noise normalization, and approximation-to-decoding inequalities all change when scales are combined.

11.2 The unresolved joint rate

The expression

$$\min\{n, kL/\varepsilon + k/\varepsilon^2\}$$

is a lower bound, not an established characterization. A matching upper bound would need to use information across levels more efficiently than the separate perforated regressions analyzed here. The two-stage algorithm reduces the final fitting cost by sharing a single fresh regression across all levels. Its pilot still uses separate perforated sketches at each level, giving the kL^2/ε term. The final rank-truncation analysis has a kL/ε^2 term because each column’s residual enters $L - 1$ blockwise noise sums. Together these leave only a single factor L between the proposed upper and lower expressions, but that factor is not a universal constant.

Proposition 11.1 (An obstruction to a uniform-embedding shortcut). *Let $m = n/2$, $\Omega \in \mathbb{R}^{n \times s}$, and $\Psi \in \mathbb{R}^{n \times t}$, where $s, t < m$. There is a nonzero rank-one $H \in \mathcal{H}_{n,k}$ satisfying $H\Omega = 0$ and $\Psi^\top H = 0$.*

Proof. Let I, J be the left and right root children. Choose unit vectors $u \in \ker(\Psi_I^\top)$ and $v \in \ker(\Omega_J^\top)$. These nullspaces are nontrivial by the width assumptions. Put $uv^\top$ in block (I, J) and zero elsewhere. This matrix has Frobenius norm one, lies in $\mathcal{H}_{n,k}$ for every $k \geq 1$, and both products vanish. $\square$

This proposition rules out an isometry or lower norm bound holding simultaneously for all HODLR differences under these two short sketches. Its witness depends on the realized sketches. It is therefore *not* a lower bound against the pointwise randomized guarantee in (1), and does not rule out a successful joint estimator with a different analysis.

Replacing the missing argument by an assertion that a joint least-squares fit must be better would not be valid: the fitted subspaces depend on the same sketch data, and rank truncation is nonlinear. Neither independence nor a uniform embedding of all possible rank-constrained differences follows from the calculations above. These are concrete gaps in that route, not steps that have been assumed in Theorems 5.1 and 7.3.

12 Reproducibility and proof-review priorities

The package includes a dense, query-counted NumPy reference implementation. Its main routine uses only an oracle’s dimension and vector-product method; it does not read the oracle’s stored matrix. Every column product, including transpose products, is counted. The parameterized experimental routine may use smaller widths, but those widths are explicitly distinguished from the conservative theorem parameters.

The recorded checks use a fixed seed and include the following results.

Check	Instances/draws
Deterministic rank-truncation inequality	3,600
Gaussian inverse and row-isotropy moments	12,000
Noisy Nyström estimates (four noise scales)	4,800
Hierarchical frame geometries	5
Selected-column anchor inequality	1,500
Fixed-query two-sided divergence identity	120
Parameter and constant inequalities	11,760
Oracle-level algorithm exercises	13

All checks in the recorded run passed. The largest observed hierarchical orthogonality defect was below 1.6×10^{-15} , and the fixed-query divergence identity discrepancies were below 7×10^{-16} . The inverse-moment and row-second-moment Monte Carlo relative discrepancies were approximately 0.0068 and 0.0139.

These tests do not certify a worst-case probabilistic theorem. In particular, the fixed-query identity check does not test adaptivity, finite randomly sampled frames do not establish packing entropy, and Monte Carlo averages do not prove expectation inequalities. The adaptive posterior argument, small-ball packing, all-parameter extension, and the exact-arithmetic interpretation require mathematical review.

The two-stage refinement is implemented separately in `two_stage.py`. Its additional checks use 3,000 shared-regression second-moment trials and 400 conditional proper-fit trials. The second-moment discrepancies are below 1.8 recorded Monte Carlo standard errors. Twelve end-to-end cases check proper output and oracle accounting; 840 parameter triples check the stated constant inequalities. The actual two-stage path also recovers a diagonal 1024×1024 example in 230 queries, with relative error below 3.3×10^{-16} . These are experimental-width exact-instance checks, not evidence of a uniform practical accuracy guarantee at those widths.

The separate known-space tests use 1,300 Gaussian trials and verify the expectation identity in Proposition 6.1 within the recorded Monte Carlo tolerances. They also recover a 128×128 known-space instance in 20 queries with relative error below 1.5×10^{-15} . Those tests supply spaces computed from explicit full matrices; this is not an additional general-oracle guarantee.

A further audit checks 240 sketch-dependent nullspace witnesses and 90 adaptively selected two-sided Gaussian histories, comprising 1,320 query steps. The largest discrepancy between the direct Gaussian posterior projector and its rectangular-span formula is below 4.8×10^{-14} . These finite-dimensional checks do not establish the conditional-independence claim for all measurable adaptive algorithms. Four extreme-accuracy input checks also exercise the n -query fallback through an opaque oracle, including the smallest positive IEEE double-precision value. The earlier package’s verification suite was rerun without changing its source; its recorded checks also pass.

For the algorithm, the small theorem-parameter examples use the exact n -query fallback. Separately, smaller experimental widths exercise actual peeling, including zero-tail inputs and nonzero perturbations. Their measured error ratios are reported without claiming 0.99 relative-error guarantees. A diagonal 1024×1024 example is recovered in 260 queries to numerical precision; this is only a query-count and exact-instance exercise.

Reproduce the checks from the package root with

```
OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1 \
python code/verify.py --output results/verification.json
OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1 \
python code/verify_two_stage.py --output results/two_stage.json
```

The earlier archive is retained unchanged under `prior/`. The appendix below incorporates its accuracy argument with only organizational changes and a uniform-extension conclusion. Both the new proofs and that earlier proof remain unreviewed.

A The earlier accuracy lower bound

This appendix supplies the k/ε^2 argument used in the main theorem. It retains the definitions and constants of the earlier note.

A.1 Information in adaptive shifted-GOE queries

For a symmetric matrix G , use the GOE normalization

$$G_{ij} = G_{ji} \sim N(0, 1/n) \quad (i < j), \quad G_{ii} \sim N(0, 2/n), \quad (44)$$

with independent upper-triangular entries. Its density on the vector space of symmetric matrices is proportional to $\exp(-n \|G\|_F^2 / 4)$. In particular, it is isotropic for the Frobenius inner product on that vector space.

Lemma A.1 (Adaptive transcript divergence). *Fix symmetric $A_0, S \in \mathbb{R}^{n \times n}$ with $\|S\|_{\text{op}} \leq 1$ and a real δ . Let $\mathbb{P}_S$ be the law of the transcript of any randomized adaptive algorithm using at most $q \leq n$ vector queries on*

$$A = A_0 + \delta S + G.$$

Let $\mathbb{P}_0$ be its transcript law on $A_0 + G$. Include the independent random seed in the transcript. Then

$$D_{\text{KL}}(\mathbb{P}_S \| \mathbb{P}_0) \leq \frac{n\delta^2 q}{2}. \quad (45)$$

The statement permits queries on either side of A .

Proof. Reduction to orthonormal fresh directions. All matrices in question are symmetric, so choosing the transpose side gives no additional type of measurement. Subtract the known product $A_0 v$ from each answer. If a new query vector has a component in the span of previous queries, its image on that component is already known. Removing it and normalizing the remaining vector recovers the same fresh information using one call. Queries wholly in the old span give no information and can be omitted. We may therefore work with an orthonormal sequence of fresh vectors, chosen adaptively. An algorithm that stops early can be padded with orthogonal queries. Data processing then bounds its original transcript by the padded transcript. Fix the seed for the moment.

Gaussian exposure under adaptivity. After $t - 1$ fresh directions, let V be the matrix of those directions and $\Pi = VV^\top$. For a fixed value of S , the observed answers determine $Z = GV$. The symmetric component of G determined by these observations is

$$G_{\text{obs}} = ZV^\top + VZ^\top - V(V^\top Z)V^\top. \quad (46)$$

The undetermined part is $(I - \Pi)G(I - \Pi)$. These are orthogonal subspaces of the space of symmetric matrices for the Frobenius inner product. Conditional on the history, the latter part remains a centered GOE compression on $\text{range}(I - \Pi)$, with the normalization $1/n$, not the inverse of the remaining dimension.

For completeness, this claim also holds when V is adaptive. Initially it is just the orthogonal decomposition of an isotropic Gaussian. Given the past, the next direction is fixed. In an orthonormal basis of the remaining space beginning with that direction, the new Gaussian column and the unexposed lower-right symmetric block are independent. Conditioning on the column leaves that block unchanged. Induction proves the assertion. The choice of the next vector is a function of already conditioned-on answers, so it does not impose an additional constraint on the unexplored block.

One-step divergence. Let v be the next unit vector, with $\Pi v = 0$. The new information is the component

$$y = (I - \Pi)(A - A_0)v.$$

The component in $\text{range}(\Pi)$ is already determined by the old answers and symmetry: $V^\top(A - A_0)v = ((A - A_0)V)^\top v$. At a fixed history, the conditional mean of y under the shifted law is $\delta(I - \Pi)Sv$, and under the baseline it is zero. In both cases its conditional covariance, restricted to $\text{range}(I - \Pi)$, is

$$\Sigma_v = \frac{1}{n}((I - \Pi) + vv^\top). \quad (47)$$

The variance is $2/n$ in direction v and $1/n$ in the orthogonal remaining directions. In particular, $\Sigma_v^{-1} \preceq nI$ on its support.

Two Gaussians with identical covariance and mean difference m have divergence $m^\top \Sigma_v^{-1} m / 2$, as follows directly by integrating the logarithm of their density ratio. The conditional divergence is therefore at most

$$\frac{n\delta^2}{2} \|(I - \Pi)Sv\|^2 \leq \frac{n\delta^2}{2}.$$

Applying the chain rule for relative entropy to the successive fresh answers and summing gives (45). Integrating over the seed gives the same bound for randomized algorithms, since its distribution is identical in both experiments. $\square$

Remark A.2 (Why exact arithmetic does not invalidate the bound). This is a divergence calculation for the laws of exact real-valued answers. No rounding, finite-precision hypothesis, bounded query coefficients, or nonadaptive restriction is used. Normalizing a query loses no information because its scale is known. The unexplored Gaussian compression is what prevents arbitrary measurable postprocessing from revealing the hidden parameter for free.

A.2 A local approximation-to-subspace inequality

The next lemma is the deterministic part of the construction. It handles arbitrary perturbations, nonsymmetric estimates, and estimates of rank strictly smaller than k .

Lemma A.3 (Local loss). *Let P be a rank- k orthogonal projector on $\mathbb{R}^{2k}$, and let $0 < \delta \leq 1/2$. Put*

$$C = (1 - \delta)I + 2\delta P.$$

For any $W \in \mathbb{R}^{2k \times 2k}$ and any matrix D of rank at most k , choose a rank- k orthogonal projector Q whose range contains the row space of D . Then

$$\begin{aligned} & \|C + W - D\|_F^2 - \tau_k(C + W)^2 \\ & \geq \delta \|P - Q\|_F^2 - \left(\frac{(1 + \delta)^2}{\delta} + 1 \right) \|W\|_F^2. \end{aligned} \quad (48)$$

Proof. Write $d = \|P - Q\|_F$ and $w = \|W\|_F$. Since $D = DQ$,

$$\|C + W - D\|_F^2 \geq \|(C + W)(I - Q)\|_F^2.$$

The rank- k matrix $(1 + \delta)P = CP$ is a valid competitor, so

$$\tau_k(C + W)^2 \leq \|C(I - P) + W\|_F^2.$$

Subtracting these inequalities and expanding squared norms yields

$$\|C(I - Q)\|_F^2 - \|C(I - P)\|_F^2 + 2\langle W, C(P - Q) \rangle_F - \|WQ\|_F^2. \quad (49)$$

Because $C^2 = (1 - \delta)^2 I + 4\delta P$, the clean difference is

$$\text{tr}((P - Q)C^2) = 4\delta(k - \text{tr}(PQ)) = 2\delta d^2.$$

The cross term is bounded below by $-2(1 + \delta)wd$, and the last term is at least $-w^2$. Thus (49) is bounded below by

$$2\delta d^2 - 2(1 + \delta)wd - w^2.$$

Use $2(1 + \delta)wd \leq \delta d^2 + (1 + \delta)^2 w^2 / \delta$ to obtain (48). $\square$

The projector Q can be selected measurably from D : choose a deterministic orthonormal basis of its row space, then extend it by a fixed coordinate-based Gram–Schmidt rule until its dimension is k . No optimality, symmetry, or uniqueness assumption on D is needed.

A.3 An explicit finite packing

All logarithms in this section and in the information calculation are natural. The metric on tuples of projectors is the square root of the sum of their squared Frobenius distances.

Lemma A.4 (Projector tuples). *For every pair of positive integers k, M , there is a finite collection*

$$\{(P_1^\theta, \dots, P_M^\theta) : \theta = 1, \dots, N\}$$

of rank- k orthogonal projectors on $\mathbb{R}^{2k}$ such that

$$\log N \geq \frac{Mk^2}{200} \quad (50)$$

and, whenever $\theta \neq \theta'$,

$$\sum_{j=1}^M \|P_j^\theta - P_j^{\theta'}\|_F^2 \geq \frac{Mk}{1600}. \quad (51)$$

Proof. Step 1: a single-block alphabet. Suppose first that $k \geq 4$. Let S range over sign matrices with independent entries $\pm 1/\sqrt{k}$. For fixed unit vectors x, y , the random variable $x^\top S y$ has moment-generating function at most $\exp(t^2/(2k))$, because $\cosh u \leq \exp(u^2/2)$ and $\sum_{ij} x_i^2 y_j^2 = 1$. Thus

$$\mathbb{P}\{|x^\top S y| > a\} \leq 2 \exp(-ka^2/2).$$

A $1/4$ -net of the unit sphere has size at most 9^k : take a maximal separated set and compare disjoint balls of radius $1/8$ with the enclosing ball of radius $9/8$. For nets in both variables, $\|S\|_{\text{op}} \leq 2 \max_{x,y \text{ in nets}} |x^\top S y|$. Consequently,

$$\mathbb{P}\{\|S\|_{\text{op}} > 10\} \leq 2 \exp(-(25/2 - 2 \log 9)k) < 1/2.$$

At least 2^{k^2-1} sign matrices are therefore good.

Let $h(p) = -p \log p - (1-p) \log(1-p)$. A binary Hamming ball of radius $\lfloor k^2/4 \rfloor$ has size at most $\exp(h(1/4)k^2)$. To verify the usual binomial bound, multiply each summand by $p^i(1-p)^{d-i}$ with $d = k^2$ and $p = 1/4$; for $i \leq pd$ this weight is at least $\exp(-dh(p))$, while the weighted sum is at most one. Greedy deletion of such balls from the good matrices gives a set with mutual Hamming distance greater than $k^2/4$ and log cardinality at least

$$(\log 2 - h(1/4))k^2 - \log 2 \geq k^2/20, \quad k \geq 4.$$

For each retained sign matrix set $T = S/20$, so $\|T\|_{\text{op}} \leq 1/2$, and let P_T project onto the graph of T , the column space of $[I \ T^\top]^\top$. For two matrices T, U , an orthonormal basis of the graph of U is $[I \ U^\top]^\top (I + U^\top U)^{-1/2}$; an orthonormal basis of the orthogonal complement of the graph of T is $[-T \ I]^\top (I + TT^\top)^{-1/2}$. Hence

$$\frac{1}{2} \|P_T - P_U\|_{\text{F}}^2 = \left\| (I + TT^\top)^{-1/2} (U - T) (I + U^\top U)^{-1/2} \right\|_{\text{F}}^2. \quad (52)$$

Both inverse square roots have minimum singular value at least $1/\sqrt{1+1/4}$. The Hamming separation also gives $\|T - U\|_{\text{F}}^2 > k/400$. It follows that

$$\|P_T - P_U\|_{\text{F}}^2 \geq \frac{2}{(1+1/4)^2} \|T - U\|_{\text{F}}^2 \geq k/400.$$

For $k = 1, 2, 3$, instead take the two complementary coordinate projectors. Their squared distance is $2k$ and their log cardinality is $\log 2 \geq k^2/20$. Thus for every k there is a single-block alphabet of size $Q \geq 2$ with $\log Q \geq k^2/20$ and squared separation at least $k/400$.

Step 2: a product code. In the Q -ary cube of length M , a Hamming ball of radius $\lfloor M/4 \rfloor$ has size at most

$$\exp(Mh(1/4) + (M/4) \log(Q-1)).$$

Indeed, factor $(Q-1)^{M/4}$ out of the relevant binomial sum and apply the binary bound just proved. Greedy packing then gives mutual Hamming distance greater than $M/4$ and

$$\log N \geq M(\log Q - h(1/4) - \frac{1}{4} \log(Q-1)) \geq \frac{1}{10} M \log Q \geq Mk^2/200.$$

The middle inequality holds for $Q \geq 2$: the function $0.9 \log Q - 0.25 \log(Q-1)$ is increasing there, and its value at 2 is greater than $h(1/4)$. Each differing coordinate contributes at least $k/400$ to squared projector distance. This proves (51). $\square$

A.4 The hard distribution and the accuracy lower bound

A.4.1 Matrices indexed by the packing

Set $M = n/(4k) = 2^{L-2}$. Partition the coordinates into M consecutive nodes of size $4k$, and split each into two sets I_j, J_j of size $2k$. These are actual sibling blocks of the HODLR partition. Crucially, they are not the unrestricted $k \times k$ leaf blocks.

Take the packing from Lemma A.4, and put

$$\begin{aligned} R_j^\theta &= 2P_j^\theta - I_{2k}, \\ A_0 &= \text{blockdiag}_{j=1}^M \begin{bmatrix} 0 & I_{2k} \\ I_{2k} & 0 \end{bmatrix}, \\ S_\theta &= \text{blockdiag}_{j=1}^M \begin{bmatrix} 0 & R_j^\theta \\ R_j^\theta & 0 \end{bmatrix}, \quad A = A_0 + \delta S_\theta + G. \end{aligned} \quad (53)$$

Each R_j^θ is a symmetric orthogonal matrix. Thus S_θ is symmetric orthogonal too, and $\|S_\theta\|_{\text{op}} = 1$. The upper targeted block of the clean matrix is

$$C_j = (1 - \delta)I_{2k} + 2\delta P_j^\theta.$$

Its singular values are $1 + \delta$ and $1 - \delta$, each repeated k times. The lower targeted block is its transpose. All other clean blocks are zero. By Lemma 2.1,

$$\text{OPT}(A_0 + \delta S_\theta)^2 = 2Mk(1 - \delta)^2 = \frac{n}{2}(1 - \delta)^2. \quad (54)$$

The packing parameters become

$$\log N \geq \frac{nk}{800}, \quad \sum_j \|P_j^\theta - P_j^{\theta'}\|_F^2 \geq \frac{n}{6400}. \quad (55)$$

A.4.2 A good-noise event

Define

$$Z = \|G\|_F^2, \quad T = \sum_{j=1}^M \|G_{I_j, J_j}\|_F^2.$$

The GOE normalization gives

$$\mathbb{E}Z = n + 1 \leq \frac{5}{4}n, \quad \mathbb{E}T = k.$$

For the second identity, there are $M(2k)^2 = nk$ entries in the targeted *upper* blocks, each with variance $1/n$. These entries are disjoint. Markov's inequality and a union bound imply that the event

$$\mathcal{E} = \{Z \leq 10n, T \leq 10k\} \quad (56)$$

has probability at least

$$\mathbb{P}(\mathcal{E}) \geq 1 - \frac{1}{8} - \frac{1}{10} = \frac{31}{40}. \quad (57)$$

On this event, use the best clean HODLR approximation as a competitor for the noisy matrix and apply $\|X + Y\|_F^2 \leq 2\|X\|_F^2 + 2\|Y\|_F^2$:

$$\text{OPT}(A)^2 \leq n(1 - \delta)^2 + 2Z \leq 21n. \quad (58)$$

No independence between Z and T is needed.

A.4.3 Decoding the hidden projectors

Suppose an approximation algorithm succeeds on an input from (53). Choose Q_j to be a rank- k projector containing the row space of B_{I_j, J_j} , using the measurable rule following Lemma A.3. Let

$$r^2 = \sum_{j=1}^M \|P_j^\theta - Q_j\|_F^2.$$

Since $\varepsilon < 1/2$, success and (58) imply, on $\mathcal{E}$,

$$\begin{aligned} \|A - B\|_F^2 - \text{OPT}(A)^2 &\leq (2\varepsilon + \varepsilon^2) \text{OPT}(A)^2 \\ &\leq \frac{105}{2}\varepsilon n. \end{aligned} \quad (59)$$

Sum Lemma A.3 over the targeted upper blocks. Their excesses are nonnegative and sum to at most the global excess by Lemma 2.1. Also,

$$\frac{(1 + \delta)^2}{\delta} + 1 = \frac{1 + 3\delta + \delta^2}{\delta} \leq \frac{3}{\delta}, \quad 0 < \delta \leq 1/2.$$

It follows that

$$r^2 \leq \frac{105\varepsilon n}{2\delta} + \frac{30k}{\delta^2}. \quad (60)$$

Now set

$$D = 2^{24}, \quad \varepsilon_0 = \frac{1}{2D}, \quad \delta = D\varepsilon, \quad (61)$$

and first restrict to

$$0 < \varepsilon \leq \varepsilon_0, \quad n \geq k/\varepsilon^2. \quad (62)$$

Then $0 < \delta \leq 1/2$ and

$$\frac{r^2}{n} \leq \frac{105}{2D} + \frac{30}{D^2} < \frac{1}{102400} < \frac{1}{25600}. \quad (63)$$

The smallest squared separation in (55) is at least $n/6400$. Thus r is less than half the minimum separation. Nearest-neighbor decoding of the tuple $(Q_1, \dots, Q_M)$ in the finite packing recovers θ uniquely. This is a measurable operation, and no query is needed to do it.

Let θ be uniform on the packing, independently of G and the algorithm's seed. The fixed-input guarantee (1) implies average success at least 0.99 under this distribution. The success event and $\mathcal{E}$ therefore intersect with probability at least

$$\frac{31}{40} - \frac{1}{100} = \frac{153}{200}. \quad (64)$$

Hence the decoding error probability is at most $47/200$.

A.4.4 Information cannot support that decoding probability

Let $\mathcal{T}$ denote the transcript including the seed. By Lemma A.1, and using the baseline transcript law generated by $A_0 + G$,

$$I(\theta; \mathcal{T}) \leq \frac{1}{N} \sum_{\theta=1}^N D_{\text{KL}}(\mathbb{P}_\theta \| \mathbb{P}_0) \leq \frac{n\delta^2 q}{2}. \quad (65)$$

The first inequality follows from the identity

$$\frac{1}{N} \sum_{\theta} D_{\text{KL}}(\mathbb{P}_\theta \| \mathbb{P}_0) = I(\theta; \mathcal{T}) + D_{\text{KL}}(\bar{\mathbb{P}} \| \mathbb{P}_0),$$

where $\bar{\mathbb{P}}$ is the mixture transcript law.

We recall the needed entropy bound directly. If $\hat{\theta}$ is any transcript-based decoder and $p_e = \mathbb{P}(\hat{\theta} \neq \theta)$, conditioning also on the error indicator gives

$$H(\theta | \mathcal{T}) \leq \log 2 + p_e \log N.$$

Since $H(\theta) = \log N$, this yields Fano's inequality in the form

$$p_e \geq 1 - \frac{I(\theta; \mathcal{T}) + \log 2}{\log N}. \quad (66)$$

Assume, for a contradiction, that

$$q \leq \frac{k}{6400\delta^2}. \quad (67)$$

This bound is less than n under (62), so the hypothesis $q \leq n$ in Lemma A.1 applies. Equations (65) and (55) give $I(\theta; \mathcal{T}) \leq (\log N)/16$. Moreover, (62) implies $nk \geq 4D^2$, so $\log N \geq nk/800 \geq 16 \log 2$. Equation (66) therefore gives $p_e \geq 7/8$, contradicting $p_e \leq 47/200$ from (64). We have proved the following.

Proposition A.5 (Accuracy lower bound in the core regime). *If $\varepsilon \leq \varepsilon_0$ and $n \geq k/\varepsilon^2$, then*

$$q_*(n, k, \varepsilon) \geq c_* \frac{k}{\varepsilon^2}, \quad c_* = (6400D^2)^{-1}. \quad (68)$$

A.5 Extending the accuracy lower bound to all parameters

Recall $D = 2^{24}$, $\varepsilon_0 = 1/(2D)$, and $c_* = (6400D^2)^{-1}$. The function $q_*(n, k, \varepsilon)$ is nonincreasing in ε . For $\varepsilon \geq \varepsilon_0$,

$$c_* \min\{n, k/\varepsilon^2\} \leq (c_*/\varepsilon_0^2)k = k/1600,$$

so Proposition 2.3 suffices. For $\varepsilon < \varepsilon_0$ and $n \geq k/\varepsilon^2$, apply Proposition A.5. Otherwise put $\zeta = \sqrt{k/n} > \varepsilon$. If $\zeta \leq \varepsilon_0$, monotonicity and the core bound at accuracy ζ give $q_* \geq c_*k/\zeta^2 = c_*n$. If $\zeta > \varepsilon_0$, then $n < k/\varepsilon_0^2$, and the exact-recovery obstruction again gives $q_* \geq c_*n$. This proves

$$q_*(n, k, \varepsilon) \geq c_* \min\{n, k/\varepsilon^2\}$$

for every parameter triple in RE-03. Together with the n -query baseline, it also proves Corollary 1.2.

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